

1 Solution — GIAM 2.3

Problem 4

1.1 Problem

Find the negation of each of the following and simplify as much as possible.

- (a) $(A \vee B) \iff C$
- (b) $(A \vee B) \implies (A \wedge B)$

1.2 Answers

$$\text{(a)} \quad \neg[(A \vee B) \iff C] \cong ((A \vee B) \wedge \neg C) \vee (\neg A \wedge \neg B \wedge C)$$

$$\text{(b)} \quad \neg[(A \vee B) \implies (A \wedge B)] \cong (A \vee B) \wedge (\neg A \vee \neg B) \cong A \oplus B$$

1.3 The Tools

Three rules do all the work — the first two from Section 2.2, the third being De Morgan:

$$\neg(P \iff Q) \cong (P \wedge \neg Q) \vee (\neg P \wedge Q),$$

$$\neg(P \implies Q) \cong P \wedge \neg Q,$$

$$\neg(P \vee Q) \cong \neg P \wedge \neg Q.$$

Push the negation inward with the standard rules, then simplify. The truth table confirms each result.

(a) $\neg[(A \vee B) \Leftrightarrow C]$		A	B	C	$(A \vee B) \Leftrightarrow C$	negation
$\neg[(A \vee B) \Leftrightarrow C]$		T	T	T	T	F
$\downarrow$ the negation of a biconditional is exclusive-or		T	T	F	F	T
$((A \vee B) \wedge \neg C) \vee (\neg(A \vee B) \wedge C)$		T	F	T	T	F
$\downarrow$ De Morgan on $\neg(A \vee B)$		T	F	F	F	T
$((A \vee B) \wedge \neg C) \vee (\neg A \wedge \neg B \wedge C)$		F	T	T	T	F
		F	T	F	F	T
		F	F	T	F	T
		F	F	F	T	F
In words: exactly one of "A or B" and "C" holds.						
Fully expanded (DNF):						
$(A \wedge \neg C) \vee (B \wedge \neg C) \vee (\neg A \wedge \neg B \wedge C)$						
		the boxed formula reproduces the "negation" column				

(b) $\neg[(A \vee B) \Rightarrow (A \wedge B)]$		A	B	$(A \vee B) \Rightarrow (A \wedge B)$	negation
$\neg[(A \vee B) \Rightarrow (A \wedge B)]$		T	T	T	F
$\downarrow \neg(P \Rightarrow Q) = P \wedge \neg Q$		T	F	F	T
$(A \vee B) \wedge \neg(A \wedge B)$		F	T	F	T
$\downarrow$ De Morgan on $\neg(A \wedge B)$		F	F	T	F
$(A \vee B) \wedge (\neg A \vee \neg B) = A \oplus B$					
		T, F, F, T is $A \Leftrightarrow B$; its negation F, T, T, F is $A \oplus B$			
"A or B, but not both" — exactly the exclusive-or of Problem 2.1.1.					
Bonus: the original statement is just $A \Leftrightarrow B$ (see the table).					

Figure 1.1: Two panels. Panel (a): starting from $\neg[(A \text{ or } B) \text{ iff } C]$, apply the rule that the negation of a biconditional is exclusive-or, giving $((A \text{ or } B) \text{ and not-}C) \text{ or } (\text{not}(A \text{ or } B) \text{ and } C)$; then De Morgan on $\text{not}(A \text{ or } B)$ gives the boxed result $((A \text{ or } B) \text{ and not-}C) \text{ or } (\text{not-}A \text{ and not-}B \text{ and } C)$. In words, exactly one of ‘A or B’ and ‘C’ holds. Fully expanded in disjunctive normal form it is $(A \text{ and not-}C) \text{ or } (B \text{ and not-}C) \text{ or } (\text{not-}A \text{ and not-}B \text{ and } C)$. An eight-row truth table lists $(A \text{ or } B) \text{ iff } C$ and its negation, confirming the formula. Panel (b): starting from $\neg[(A \text{ or } B) \text{ implies } (A \text{ and } B)]$, apply $\neg(P \text{ implies } Q) \text{ equals } P \text{ and not-}Q$, giving $(A \text{ or } B) \text{ and not}(A \text{ and } B)$; then De Morgan gives the boxed result $(A \text{ or } B) \text{ and } (\text{not-}A \text{ or not-}B)$, which equals $A \text{ xor } B$ — ‘A or B but not both’, exactly the exclusive-or of Problem 2.1.1. A four-row truth table shows the original column is T, F, F, T, which is $A \text{ iff } B$, and its negation F, T, T, F is $A \text{ xor } B$.

1.4 Part (a)

Write $P = A \vee B$. The negation of a biconditional says the two sides *differ*:

$$\neg(P \Leftrightarrow C) \cong (P \wedge \neg C) \vee (\neg P \wedge C).$$

Substituting $P = A \vee B$ and applying De Morgan to $\neg(A \vee B) = \neg A \wedge \neg B$:

$$\neg[(A \vee B) \Leftrightarrow C] \cong ((A \vee B) \wedge \neg C) \vee (\neg A \wedge \neg B \wedge C).$$

In words: **exactly one** of " A or B " and " C " holds. Distributing the first term gives the fully expanded disjunctive normal form

$$(A \wedge \neg C) \vee (B \wedge \neg C) \vee (\neg A \wedge \neg B \wedge C),$$

though the two-term version above is the more readable answer.

1.5 Part (b)

Here the negation of a conditional applies: $\neg(P \Rightarrow Q) \cong P \wedge \neg Q$, with $P = A \vee B$ and $Q = A \wedge B$:

$$\neg[(A \vee B) \Rightarrow (A \wedge B)] \cong (A \vee B) \wedge \neg(A \wedge B).$$

De Morgan on $\neg(A \wedge B)$ gives the clean form

$$(A \vee B) \wedge (\neg A \vee \neg B),$$

which reads " A or B , but not both" — precisely the **exclusive-or** $A \oplus B$. Equivalently, in disjunctive normal form, $(A \wedge \neg B) \vee (\neg A \wedge B)$.

1.6 Remark — Part (b) Comes Full Circle to the XOR Circuit

The answer to (b) is exactly the formula built in the very first problem of the chapter. Problem 2.1.1 constructed a circuit for $A \oplus B$ from the sum-of-products form $(A \wedge \neg B) \vee (\neg A \wedge B)$, and noted in passing the equivalent “or but not both” form $(A \vee B) \wedge \neg(A \wedge B)$. That second form is precisely what falls out here. So the negation of “if either holds then both hold” *is* the XOR gate — and the circuit drawn back in 2.1.1 already computes it.

1.7 Remark — The Original Statement in (b) Is Just $A \Leftrightarrow B$

There is a hidden simplification lurking on the *other* side. The truth table of $(A \vee B) \Rightarrow (A \wedge B)$ is T, F, F, T — which is the table of $A \Leftrightarrow B$. Indeed the statement

says “if at least one holds, then both hold,” which forces A and B to agree; and when neither holds they agree vacuously. So

$$(A \vee B) \Rightarrow (A \wedge B) \cong A \Leftrightarrow B.$$

That makes part (b) an instance of the general fact $\neg(A \Leftrightarrow B) \cong A \oplus B$ (the last row of the negation table in Problem 2.2.4). Spotting this first would have given the answer in one step — a good illustration of the habit from Problem 2: **simplify before you negate**, since a simpler starting formula usually yields a simpler negation.

1.8 Remark — Negation Never Simplifies a Biconditional Away

Contrast the two parts. In (b) the negation *collapsed* — three connectives became one XOR. In (a) it did not: the answer still needs two clauses, because a biconditional genuinely relates three independent letters here and there is no shorter equivalent. This is worth noticing because “simplify as much as possible” does not promise a one-symbol answer; it asks for the shortest *correct* form. A useful stopping test is whether every remaining clause is doing real work — in (a), dropping either disjunct changes the truth table, so both are needed.

1.9 Remark — Why Negating a Biconditional Gives XOR

The rule used in (a) deserves a one-line justification. $P \Leftrightarrow Q$ is true exactly when P and Q have the *same* truth value; so its negation is true exactly when they **differ** — and “differ” is the definition of exclusive-or. Symbolically,

$$\neg(P \Leftrightarrow Q) \cong P \oplus Q \cong (P \wedge \neg Q) \vee (\neg P \wedge Q).$$

This is the reason $\Leftrightarrow$ and $\oplus$ are sometimes called **XNOR** and **XOR** in hardware: each is the other’s negation. As noted in Problem 2.1.1, XOR is also addition modulo 2, so a biconditional is “the two bits are equal” and its negation is “the two bits sum to 1.”

1.10 Remark — A General Strategy for Negating Compound Statements

Both parts follow one algorithm, worth stating explicitly:

1. **Identify the outermost connective** and apply its negation rule ($\Leftrightarrow \rightarrow \oplus$, $\Rightarrow \rightarrow P \wedge \neg Q$, $\wedge \rightarrow \vee$ via De Morgan, $\vee \rightarrow \wedge$).
2. **Push the remaining negations inward**, repeating step 1 on each sub-formula, until every $\neg$ sits directly on a variable.
3. **Simplify** using the absorption, distributive, and complement laws of Problems 1–2.

Step 2's end state — negations only on letters — is called **negation normal form**, and it is the standard preprocessing step before converting to CNF or DNF. Working outside-in is what keeps a multi-layer negation from turning into guesswork; each rule application strips exactly one layer.